

At the time of writing in 2013, the estate tax exemption is \$5.25 million per spouse, or \$10.5 million per married couple.²² The top tax rate assessed for estates in excess of these amounts is 40%. Ultra high net worth asset owners have many trusts they can play with: grantor retained annuity trusts (GRATs), charitable remainder unitrusts, and their close cousins charitable lead trusts, intentionally defective grantor trusts (IDGTs, which are called “I Dig It” trusts), and dynasty trusts—just to name a few. GRATs are very popular among the wealthy and it would be rare to find an ultra-high net worth asset owner without at least one GRAT in her estate. Along with these instruments are family partnerships or corporate structures with valuation and control discounts, which set a different value of the shares for tax purposes than more realistic market values.

While the rules for these are very complicated (that is why you hire your tax attorney), they have three key economic components:

1. Alternative location

They offer taxpayers another asset location with a tax break. GRATs are initially set up by a donation to a tax-advantaged trust.

2. The trust generates income

The trust takes advantage of income compounding at tax-advantaged rates (see Section 1). In a GRAT, the trust pays money to the person setting up the trust, which is taxable, but any appreciation in excess of a hurdle rate (which is computed by the IRS, is adjusted every month, and is based on a market interest rate) flows back into the trust. Thus assets with high expected returns are ideally held in GRATs. They compound tax free.

3. The trust transfers to beneficiaries

At death, assets in the trust are transferred to beneficiaries and are not subject to the gift tax. (In 2012, the IRS permitted a parent to transfer up to \$5.12 million to a child tax free.) The GRAT is set up with a specific maturity (at least two years), and one risk is that if the grantor dies before the trust matures, then the value of the trust is included in her estate—but this would have been exactly the same situation if the GRAT had not been set up, so the grantor’s estate has not lost anything (except the lawyer’s fees in setting up the GRAT).

Absent the probability of dying before the expiration of the trust, computing optimal allocations to a GRAT can be handled in Huang’s model. With appropriate assumptions on the expected returns and dividend streams of the assets, we can

²²An exemption of \$5 million was introduced in 2010 under the Tax Relief, Unemployment Insurance Reauthorization, and Job Creation Act (TRA), but at the time of passage the exemption was temporary. In 2013, the American Taxpayer Relief Act made the TRA exemption permanent and indexed it to inflation.